INCH-POUND

MIL-A-52019C(ME)
28 June 1991
SUPERSEDING
MIL-A-52019B(ME)
22 September 1981

MILITARY SPECIFICATION

ADAPTER ASSEMBLY, GLIDE ANGLE LIGHT TO STAKE OR CONE

This specification is approved for use within the USA Belvoir Research Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an adapter assembly for airport and heliport glide angle lights.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6210

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SPECIFICATIONS

FEDERAL

HH-L-361	- Lining, Brake; and Lining Material, Brake: Automotive.
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Government drawing. The following Government drawing forms a part of this specification to the extent specified. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

ME

TA13205E3550	- Electric Lighting Equipment Sets Airport and Heliport, Adapter Assembly Glide Angle Light.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FGS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise

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specified, the issues of documents not listed in the DODISS are the issues of the non-Government document cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 26	- Aluminum-Alloy Sand Castings, Specification for.
B 209	- Aluminum and Aluminum-Alloy Sheet and Plate, Specification for.
B 211	- Aluminum-Alloy Bar, Rod, and Wire, Specification for.
B 221	- Aluminum-Alloy Extruded Bars, Rods, Wire, Shapes, and Tubes, Specification for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The mounting adapter shall be constructed in accordance with TA13205E3550, and as specified herein. The adapter shall rotate freely through 360 degrees, and all adjusting screws and bolts shall turn freely by hand throughout their range of adjustment. Thumb screws and knurled bolts shall securely lock the moving parts of the adapter, and the adapter shall securely mount on a 1.25-inch by 1.25-inch by 0.25-inch (3.18-cm by 3.18-cm by 0.64-cm) standard aluminum angle when adjustment screws are tightened by hand.

3.1.1 Drawings. The drawings forming a part of this specification are end-product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the mounting adapter. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

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3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Aluminum alloy. Aluminum alloy shall conform to the following specifications unless otherwise shown on the drawings:

Form	Specifications
Castings, sand	ASTM B 26, alloy 443.0 temper F
Sheets and plates	ASTM B 209, alloy 6061, temper T6
Bar, rods, and shapes	ASTM B 211, alloy 6061, temper T6; or ASTM B 221, alloy 6061, temper T6

3.3.2 Friction disc material. Friction disc material shall conform to HH-L-361, type II, composition B, class M.

3.3.3 Material deterioration prevention and control. The adapter shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the adapter may be exposed.

3.3.3.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the adapter may be newly fabricated from recovered materials to the maximum extent practicable, provided the adapter produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the adapter.

3.4 Treatment and painting. The portions of the adapter normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable, color green 383.

3.5 Identification marking. The adapter shall be identified in accordance with MIL-STD-130.

3.6 Workmanship. All parts and components of the mounting adapter including castings, stampings, machined surfaces, and welds shall be clean and

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free from sand, dirt, fins, pits, sprues, scale, flux, weld splatter, or any other defects that could impair its operation or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The mounting adapter shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II, table I and table IIa.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for major defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

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4.5.1 Examination. The mounting adapters shall be examined for the following defects:

Number	Defects	Requirement paragraph
Major		
101.	Nonconformance with any attribute designated for inspection by the drawings.	3.1
102.	Adapter does not rotate freely through 360 degrees.	3.1
103.	Any adjusting screw or bolt does not turn freely by hand throughout its range of adjustment.	3.1
104.	When clamped on a 1.25-inch by 1.25-inch by 0.25-inch (3.18-cm by 3.18-cm by 0.64-cm) standard aluminum angle and adjustment screws are tightened by hand, any relative motion between the adapter and the angle can be felt by hand.	3.1
105.	Thumb screws and knurled bolts do not securely lock moving parts of adapter.	3.1
106.	Materials not as specified.	3.3
107.	Aluminum alloy not as specified.	3.3.1
108.	Friction disc material not as specified.	3.3.2
109.	Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.3
110.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.3.1
111.	Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.3.3.2
112.	Used, rebuilt or remanufactured components, pieces, or parts are incorporated in the kit.	3.3.4
113.	Treatment and painting not as specified.	3.4
114.	Identification marking missing, incomplete or not legible.	3.5
115.	Workmanship not as specified.	3.6

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

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4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, table I and table IIa.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

Number	Defects	Requirement paragraph
Major		
116.	Box closure and sealing not in accordance with the specified method for level A.	5.1.1
117.	Shipping containers not as specified for levels A, B, or C.	5.2.1, 5.2.2, 5.2.3
118.	Quantities packed together exceed specified quantities for level A or B.	5.2.1, 5.2.2
119.	Marking missing, illegible, incorrect or incomplete.	5.3

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each adapter assembly shall be preserved in accordance with MIL-P-116, method III using a close-fitting box conforming to PPP-B-636, W6c, style optional. Box closure and sealing shall be as specified for method V in the appendix to the box specification.

5.1.2 Level C. Adapter assemblies shall be placed in a close-fitting box conforming to PPP-B-636, class domestic, grade as appropriate for the size and weight of the contents. Cushioning shall be provided to prevent movement and damage. Box closure shall be in accordance with the box specification.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Adapter assemblies, preserved as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed 200 pounds. Strapping shall be in accordance with QQ-S-781, class 1, type I or IV, finish B, size as applicable. Box closure and strapping shall be in accordance with the appendix to the box specification.

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5.2.2 Level B. Adapter assemblies, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-636, V3c, style optional, or PPP-B-640, class 2, grade A, style optional. The gross weight or size shall not exceed the limitations of the box specification. Box closure and sealing shall be in accordance with the appendix to the applicable box specifications.

5.2.3 Level C. Adapter assemblies, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-636, domestic class. Box closure and sealing shall be in accordance with the appendix to the applicable box specification.

5.3 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The adapters are intended for use in mounting a glide angle approach light to a stake or cone.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article will not be required (see 3.2).
d. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 4.1).
e. Level of preservation and packing required (see 5.1 and 5.2).
f. Any special marking (see 5.3).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior

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Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (key word) listing.

Airport
Heliport
Lights, glide angle

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project: 6210-A639

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-A-52019C(ME)	91/06/25

3. DOCUMENT TITLE

Adapter Assembly, Glide Angle Light to Stake or Cone

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466	

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